
PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

May/June 2019

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **9** printed pages.



Cambridge Assessment
International Education

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1	Defining the problem	
	m is the independent variable and θ is the dependent variable or vary m and measure θ or θ is the independent variable and m is the dependent variable or vary θ and measure m	1
	keep position of load <u>constant</u> or L <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - load shown touching at P - load labelled and at least one other label	1
	use a protractor to measure θ	1
	if m is the independent variable: (slowly) change the angle until the block (just) topples or if θ is the independent variable: (slowly) change mass until block (just) topples	1
	use a balance to measure m	1

Question	Answer	Marks
1	Method of analysis	
	plot a graph of $\cos \theta$ against $1/m$ (or reverse axes $1/m$ against $\cos \theta$) (not $1/\cos \theta$ against m or m against $1/\cos \theta$)	1
	relationship valid if a straight line (not ‘through the origin’ unless the choice of axes indicates the intercept is zero)	1
	$\alpha = \frac{2L \times \text{gradient}}{V \times w} \quad \text{or} \quad \alpha = \frac{\text{gradient}}{V \times y\text{-intercept}}$ for reverse axes: $\alpha = \frac{2L}{V \times w \times \text{gradient}} \quad \text{or} \quad \alpha = -\frac{1}{V \times y\text{-intercept}}$	1

Question	Answer	Marks
1	Additional detail including safety considerations	Max. 6
D1	use cushion/foam/sandbox <u>in case block/load falls</u>	
D2	measure L with a rule	
D3	correctly positioned protractor to measure θ , e.g. protractor with its centre over the nail and its straight edge parallel to an edge of the block	
D4	method to fix m to strip	
D5	method to determine volume of block, e.g. $V = w \times h \times l$	
D6	measure w , h and l with calipers/micrometer/rule	
D7	(if m is the independent variable:) repeat experiment for θ <u>and</u> determine the average θ or (if θ is the independent variable:) repeat experiment for m <u>and</u> determine the average m	
D8	equation must be in the form of $y = mx + c$ with $\cos \theta$ on one side and m on the other side e.g. for a graph of $\cos \theta$ against $1/m$ $\cos \theta = \frac{\alpha V w}{2 L m} + \frac{w}{2 L}$ or for a graph of $1/m$ against $\cos \theta$ $\frac{1}{m} = \frac{2 L \cos \theta}{\alpha V w} - \frac{1}{\alpha V}$	
D9	method to determine centre of load or centre of block e.g. measure diameter/width and halve or diagonals across the block	
D10	method to ensure that block or strip is horizontal, e.g. check with a spirit level that table/block is horizontal or use a rigid strip	

Question	Answer		Marks														
2(a)	gradient = $4\pi^2 M$		1														
2(b)	<table><tr><th>T / s</th><th>T^2 / s^2</th></tr><tr><td>1.12 or 1.120</td><td>1.25 or 1.254</td></tr><tr><td>0.950 or 0.9500</td><td>0.903 or 0.9025</td></tr><tr><td>0.815 or 0.8150</td><td>0.664 or 0.6642</td></tr><tr><td>0.655 or 0.6550</td><td>0.429 or 0.4290</td></tr><tr><td>0.570 or 0.5700</td><td>0.325 or 0.3249</td></tr><tr><td>0.47 or 0.470</td><td>0.22 or 0.221</td></tr></table>		T / s	T^2 / s^2	1.12 or 1.120	1.25 or 1.254	0.950 or 0.9500	0.903 or 0.9025	0.815 or 0.8150	0.664 or 0.6642	0.655 or 0.6550	0.429 or 0.4290	0.570 or 0.5700	0.325 or 0.3249	0.47 or 0.470	0.22 or 0.221	
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	Values of T as above.		1														
	Values of T^2 as above.		1														
	Uncertainties in T increase from ± 0.01 to ± 0.02 .		1														
	Uncertainties in T^2 about ± 0.02 .		1														
2(c)(i)	Six points plotted correctly. Must be accurate to the nearest half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in T^2 plotted correctly. All error bars to be plotted. Length of bar must be accurate to less than half a small square and symmetrical.		1														
2(c)(ii)	Line of best fit drawn. If points are plotted correctly then lower end of line should pass between (0.048, 0.5) and (0.052, 0.5) and upper end of line should pass between (0.098, 1.0) and (0.104, 1.0).		1														
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.		1														

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Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of points from the line of best fit into $\Delta y / \Delta x$. Distance between points must be at least half the length of the drawn line.	1
	uncertainty = gradient of line of best fit – gradient of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)(i)	M determined from gradient and given to 2 or 3 significant figures and with correct unit. $M = \frac{\text{gradient}}{4\pi^2} = \frac{\text{(c)(iii)}}{39.478}$	1
2(d)(ii)	% uncertainty in M = % uncertainty in gradient	1

Question	Answer	Marks
2(e)	k calculated. Correct substitution of numbers required. $k = \left(\frac{4\pi^2 M}{T^2} \right) = \frac{4\pi^2 (\text{d})(\text{i})}{2.5^2} \text{ or } 6.3165 \times (\text{d})(\text{i})$ or $k = \left(\frac{\text{gradient}}{T^2} \right) = \frac{(\text{c})(\text{iii})}{2.5^2} \text{ or } \frac{(\text{c})(\text{iii})}{6.25}$	1
	Absolute uncertainty in k. Correct substitution of numbers required. Using M: $\text{uncertainty in } k = \left(\frac{\Delta M}{M} + 2 \times \frac{\Delta T}{T} \right) \times k$ $\text{uncertainty in } k = \left(\frac{(\text{d})(\text{ii})}{100} + 0.008 \right) \times k$ $\max k = \frac{4\pi^2 \times \max M}{\min T^2} \text{ or } \min k = \frac{4\pi^2 \times \min M}{\max T^2}$ Using gradient: $\text{uncertainty in } k = \left(\frac{\Delta \text{gradient}}{\text{gradient}} + 0.008 \right) \times k$ $\max k = \frac{\max \text{gradient}}{\min T^2} \text{ or } \min k = \frac{\min \text{gradient}}{\max T^2}$	1